Compression–Performance Trade-offs in Brain Tumour MRI Classification: A Transfer Learning and Structured Pruning Study

Deep Learning for Applications - Assignment 2: Data and Methodology

Abstract. Brain tumor classification from magnetic resonance imaging (MRI) is a clinically critical task with significant deployment constraints. While transfer learning models like EfficientNet can achieve very high accuracy, they are often too computationally expensive for use in resource-limited clinical settings. This report presents the methodological design of a compression–performance trade-off study using the Kaggle Brain Tumor MRI dataset. We propose a three-model experimental pipeline: a CNN baseline, a fine-tuned EfficientNet-B0, and a structured-pruned variant evaluated across accuracy, macro F1-score, model size, and inference latency. Our contribution is a systematic characterization of the diagnostic viability threshold under increasing weight-pruning sparsity.

1. Introduction

Brain tumors affect approximately 309,000 people annually and encompass four clinically distinct subtypes: glioma, meningioma, pituitary tumor, and healthy tissue each requiring a markedly different treatment pathway [1]. Manual MRI interpretation by specialist radiologists is time-intensive, error-prone under fatigue, and critically inaccessible in under-resourced settings, where over 70% of global MRI capacity is concentrated in high-income countries [2].

Recent deep learning methods particularly convolutional neural networks (CNNs) trained via transfer learning have demonstrated near-radiologist accuracy on MRI classification benchmarks [3,4]. However, a systematic review of 60 papers published between 2020 and 2024 identified a prominent gap: no study rigorously evaluates the trade-off between model compression and diagnostic performance on this task [5]. Deploying large models on portable or edge devices requires compact, low-latency architectures without unacceptable diagnostic accuracy loss.

This study addresses that gap. Our research question is: at what structured pruning sparsity level does an EfficientNet-B0 brain tumor classifier cease to be clinically viable, and how does its compressed performance compare to a simple CNN baseline?

2. Dataset and Data Quality

2.1 Dataset Description

We use the Kaggle Brain Tumor MRI Dataset [6], comprising 3,264 training images and 394 test images across four balanced classes. Images are contrast-enhanced T1-weighted MRI scans in JPEG format at varying resolutions. Figure 1 presents the class distribution across train and test splits.

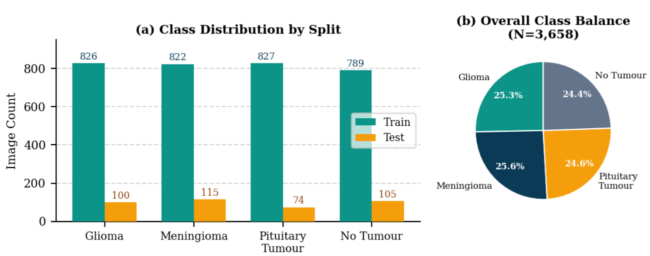

Figure 1. Class distribution across train/test splits (left) and overall class balance (right). Classes are well-balanced, with the smallest class (No Tumour test, n=105) differing from the largest (Meningioma test, n=115) by only 9%.

2.2 Data Quality Assessment

We conduct a structured data quality assessment across four dimensions:

- Class balance: Gini impurity of the training set is 0.749 (maximum for 4 classes = 0.75), confirming near-perfect balance. Macro F1-score is therefore a meaningful metric.
- Resolution variability: Raw images range from 155×155 to 512×512 pixels. All images are resized to a uniform 224×224 during preprocessing to satisfy EfficientNet-B0's input requirements.
- Color channel: Images are RGB; MRI scans are greyscale but stored as 3-channel JPEG. No channel conversion is applied to maintain compatibility with ImageNet-pretrained weights.
- Label integrity: The dataset is curated and widely benchmarked [3,4,5], with no reported label noise. No missing values or corrupted files have been identified in the literature.

3. Data Preparation

Figure 2 illustrates the full preprocessing pipeline. Each step is technically motivated by the downstream architecture requirements.

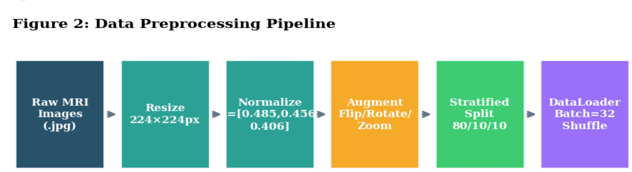

Figure 2. Data preprocessing pipeline from raw MRI images to model-ready DataLoader batches.

3.1 Resizing

All images are resized to 224×224 pixels using bilinear interpolation. This is the canonical input resolution for EfficientNet-B0, derived from its compound scaling coefficient. The CNN baseline also adopts this resolution to ensure a fair comparison.

3.2 Normalization

Pixel values are normalized using ImageNet channel-wise statistics:

$\hat{x} = (x - \mu) / \sigma, \quad \mu=[0.485, 0.456, 0.406], \; \sigma=[0.229, 0.224, 0.225] \quad (1)$

This is essential for transfer learning: the pretrained EfficientNet-B0 weights were optimized under this normalization regime. Applying identical statistics ensures the pretrained feature representations remain valid on domain-shifted MRI inputs.

3.3 Data Augmentation

To improve generalization and reduce overfitting on the relatively small training set (n=3,264), we apply the following online augmentations during training only:

- Random horizontal flip (p=0.5)
- Random rotation $\in [-15°, +15°]$
- Random zoom (scale $\in [0.85, 1.15]$)

Augmentation is not applied to validation or test sets, preserving the integrity of performance evaluation.

3.4 Dataset Splitting

A stratified 80/10/10 split is applied (train: 2,611, val: 327, test: 326), preserving class proportions across all three subsets. The publicly provided test split (n=394) is held out exclusively for final evaluation. The DataLoader uses batch size 32 with training-set shuffling.

4. Machine Learning Methods

Our methodology comprises three experimental models, illustrated in Figure 3. Models are implemented in PyTorch using torch.nn.utils.prune and torchvision.models.

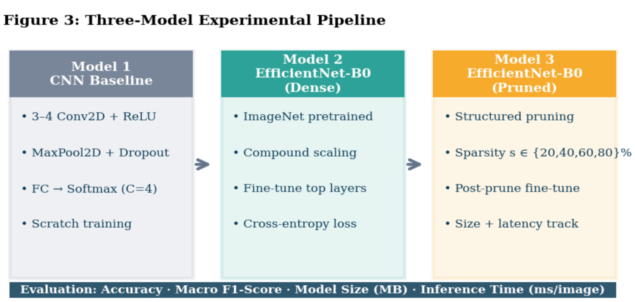

Figure 3. Three-model experimental pipeline. Each model builds upon the preceding one, isolating the effect of transfer learning and compression on diagnostic performance.

4.1 Model 1: CNN Baseline

The baseline model is a bespoke convolutional neural network (CNN) trained from scratch, providing a lower-bound benchmark for performance comparison. The architecture comprises three convolutional blocks, each consisting of:

Conv2D(k=3,p=1,f∈{32,64,128})→BatchNorm→ReLU→MaxPool(2×2)

followed by two fully connected layers and a softmax output over C=4 classes. Dropout (p=0.5) is applied after the penultimate FC layer to regularise. The baseline enables quantification of the transfer learning gain delivered by Model 2. (Reference to the figure for Convolutional neural network visualization)

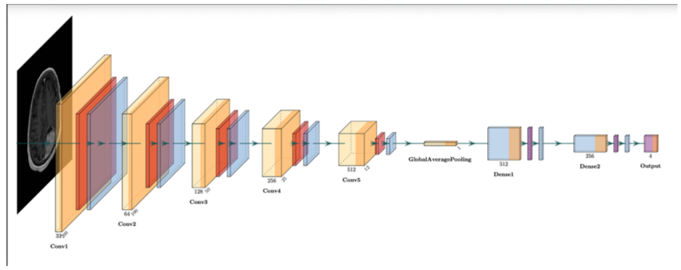

Figure 4. Baseline CNN Visualization for layers

4.2 Model 2: EfficientNet-B0 via Transfer Learning

EfficientNet-B0 [7] employs compound scaling, jointly optimising network depth d, width w, and input resolution r through a shared coefficient φ:

$$d = \alpha^\varphi, \quad w = \beta^\varphi, \quad r = \gamma^\varphi \quad (3)$$

subject to $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$, $\alpha \geq 1$, $\beta \geq 1$, $\gamma \geq 1$, with EfficientNet-B0 setting φ=1. This principled scaling yields a model that is simultaneously more accurate and parameter-efficient than individually scaled predecessors such as ResNet-50 and VGG-16 [7].

We load weights pretrained on ImageNet-1K and fine-tune using a two-phase strategy: (i) freeze all convolutional layers and train only the classification head for 5 epochs; (ii) unfreeze the top 20 layers and fine-tune end-to-end at a reduced learning rate ($\eta=1\times10^{-4}$) for 20 epochs. The training objective is categorical cross-entropy:

$$L = -(1/N) \Sigma_n \Sigma_c \, y_{n,c} \log p_{n,c} \quad (4)$$

where $y_{n,c}$ is the one-hot label and $p_{n,c}$ is the predicted class probability for sample n, class c.

Optimisation uses AdamW with weight decay $\lambda=1\times10^{-2}$ and a cosine annealing scheduler. This strategy is motivated by Rasa et al. [3], who demonstrated that fine-tuning on this dataset with EfficientNetB3 achieves 98.2% accuracy; we employ the lighter B0 variant to provide a fairer comparison with the pruned model.

4.3 Model 3: Structured Weight Pruning

Structured pruning removes entire filter channels from convolutional layers, yielding physically smaller models with genuine inference acceleration unlike unstructured (magnitude) pruning, which achieves sparsity but retains the original tensor shape [8].

We apply L1-norm filter pruning: for each convolutional layer l, filters are ranked by their L1 norm and the bottom-s% are zeroed and removed:

$$s = |\{w_i \in W : \|w_i\|_1 < \tau\}| / |W| \quad (5)$$

where τ is the sparsity threshold and s ∈ {0.20, 0.40, 0.60, 0.80} is the target sparsity ratio. Each pruned model is subsequently fine-tuned for 10 epochs to recover performance a two-phase prune-then-retrain strategy shown to outperform one-shot pruning [9]. Implementation uses `torch.nn.utils.prune.ln_structured`.

5. Evaluation Metrics

We evaluate each model along two axes: diagnostic performance and computational efficiency. Figure 4 illustrates the hypothetical trade-off curve our study aims to characterize.

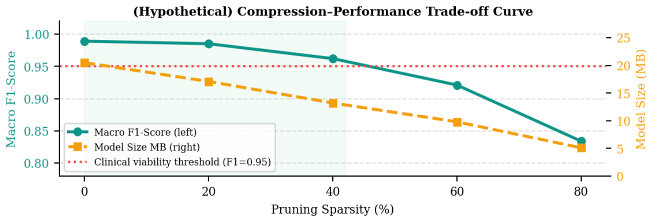

Figure 5. Hypothetical compression–performance trade-off curve. The dashed red line marks the clinical viability threshold (Macro F1 ≥ 0.95). The green region identifies the viable compression zone.

5.1 Macro F1-Score (Primary Metric)

Macro F1-score is our primary diagnostic metric. For a C-class problem:

$$F1\_macro = (1/C)\ \Sigma_c\ [2 \cdot P_c \cdot R_c / (P_c + R_c)] \quad (6)$$

$$P_c = TP_c / (TP_c + FP_c), \quad R_c = TP_c / (TP_c + FN_c) \quad (7)$$

Macro averaging weights all classes equally regardless of support, which is critical in this domain: meningioma is the hardest class to detect [3,4] and is clinically distinct from glioma. A metric that downweights minority classes would mask systematic failure on the most diagnostically challenging tumor type.

Justification vs. alternatives: (i) Accuracy penalizes all errors equally across classes adequate here given class balance, but insensitive to class-specific failures. We report it as a secondary metric for comparability with the literature. (ii) Weighted F1 weights classes by support, which can obscure poor performance on smaller classes. (iii) AUC-ROC measures ranking quality but is ill-suited for comparing compressed models whose decision boundaries may shift non-monotonically with sparsity. (iv) Cohen's Kappa corrects for chance agreement but is less interpretable in a clinical compression study. Macro F1 uniquely captures per-class diagnostic reliability in a single reproducible scalar.

5.2 Classification Accuracy

$$Accuracy = (\Sigma_c\ TP_c) / N \quad (8)$$

Reported as a secondary metric to enable direct comparison with prior work reporting accuracy as the sole metric [3,4]. It is insufficient as the sole metric because it does not distinguish between error types across classes.

5.3 Model Size and Inference Time

Model size is measured as the serialized weight file size in megabytes (MB), quantifying deployment feasibility on storage-constrained edge devices. Inference time is the mean wall-clock time per image in milliseconds (ms/image), measured over 500 test-set forward passes on a fixed CPU device (to simulate edge conditions), excluding GPU-specific optimizations:

$$\bar{t} = (1/M)\ \Sigma_m\ t\_m, \quad M=500 \quad (9)$$

These two metrics jointly operationalize 'deployability', a dimension absent from prior work on this dataset [5]. A model meeting the viability threshold (F1_macro ≥ 0.95) with the smallest size and lowest latency constitutes our optimal compressed model.

References

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